

## Lecture #02: Multivariable Calculus.

①

→ values of Function

→ Elements of Analytical Geometry

↳ 2D geometry

↳ 3D geometry

Example: If  $f(x) = 2x^2 - 1$

then  $f(1) = 2(1)^2 - 1 = 1$

$$f(4) = 2(4)^2 - 1 = 31$$

$$f(-2) = 2(-2)^2 - 1 = 7$$

$$f(t-4) = 2(t-4)^2 - 1 = 2t^2 - 16t + 31$$

Example

$$f(x, y) = x^2 y + 1$$

then  $f(2, 1) = (2)^2(1) + 1 = 5$

$$f(1, 2) = (1)^2(2) + 1 = 3$$

$$f(0, 0) = (0)^2(0) + 1 = 1$$

$$f(1, -3) = (1)^2(-3) + 1 = -2$$

$$f(3a, a) = (3a)^2(a) + 1 = 9a^3 + 1$$

$$f(ab, a-b) = (ab)^2(a-b) + 1 = a^3b^2 - a^2b^3 + 1$$

Ex. Let  $f(x, y) = x + \sqrt[3]{xy}$

(a)  $f(2, 4) = 2 + \sqrt[3]{2(4)} = 2 + \sqrt[3]{8} = 2 + \sqrt[3]{2 \times 2 \times 2}$   
 $= 2 + 2 = 4.$

(b)  $f(t, t^2) = t + \sqrt[3]{t t^2}$   
 $= t + \sqrt[3]{t^3}$   
 $= t + t = 2t$

(c)  $f(x, x^2) = x + \sqrt[3]{x \cdot x^2}$   
 $= x + \sqrt[3]{x^3}$   
 $= x + \sqrt[3]{x \cdot x \cdot x}$   
 $= x + x = 2x$

(d)  $f(2y^2, 4y) = 2y^2 + \sqrt[3]{2y^2 \cdot 4y}$   
 $= 2y^2 + \sqrt[3]{8y^3}$   
 $= 2y^2 + \sqrt[3]{2y \cdot 2y \cdot 2y}$   
 $= 2y^2 + 2y$

$$\text{let } f(x, y, z) = \sqrt{1 - x^2 - y^2 - z^2} \quad (3)$$

then

$$f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \sqrt{1 - (0)^2 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2}$$

$$= \sqrt{\frac{1}{2}}$$

Ex:  $f(x, y, z) = xyz^3 + 3$

•  $f(2, 1, 2) = 2(1)^2(2)^3 + 3 = 19$

$f(0, 0, 0) = (0)(0)^2(0)^3 + 3 = 3$

$f(a, a, a) = a(a)^2(a)^3 + 3 = a^6 + 3$

$f(t, t^2, -t) = (t)(t)^2(-t)^3 + 3 = -t^7 + 3$

•  $f(-3, 1, 1) = (-3)(1)^2(1)^3 + 3 = 0$

Ex: let  $f(x, y, z) = x^2 y^2 z^4$

$x(t) = t^3$ ,  $y(t) = t^2$  and  $z(t) = t$

then

$$f(x(t), y(t), z(t)) = (x(t))^2 (y(t))^2 (z(t))^4$$

$$= (t^3)^2 (t^2)^2 (t^4) = t^{14}$$

Ex:  $f(x, y, z) = xyz + x$

then

$$f(xy, y/x, xz) = \left[ (xy) \left( \frac{y}{x} \right) (xz) + xy \right]$$
$$= xy^2z + xy$$

Ex: let

$$g(x, y, z) = z \sin xy$$

$$u(x, y, z) = z^3 x^2$$

$$v(x, y, z) = \rho xyz$$

$$\text{and } w(x, y, z) = xy/z$$

then

$$g(u(x, y, z), v(x, y, z), w(x, y, z))$$
$$= w(x, y, z) \sin(u(x, y, z) \cdot v(x, y, z))$$
$$= \left( \frac{xy}{z} \right) \sin(x^2 z^3 \rho xyz)$$
$$= \left( \frac{xy}{z} \right) \sin(\rho x^3 y z^4)$$

$$\sqrt{\frac{x-2}{x}} \quad \frac{1}{0} = \infty, \quad \sqrt{-1} = i$$

(5)

Domain

### Function of one variable

A function  $f$  of one real variable  $x$  is a rule that assigns a unique real number  $f(x)$  to each point  $x$  in some set  $D$  of the real line.

### Function of two variables

A function  $f$  in two real variables  $x$  and  $y$  is a rule that assigns unique real number  $f(x,y)$  to each point  $(x,y)$  in some set  $D$  of the  $xy$ -plane.

### Function of three variables

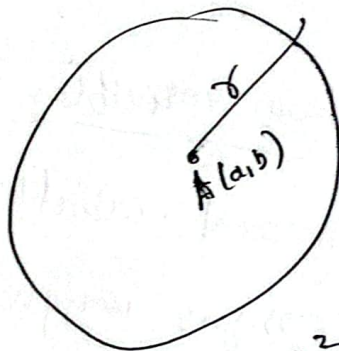
A function  $f$  in three real variables  $x, y$  and  $z$ , is a rule that assigns a unique real number  $f(x,y,z)$  to each point  $(x,y,z)$  in some set  $D$  of three dimensional space.

## Function of n variables

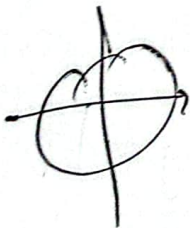
A function  $f$  in  $n$  variable real variables  $x_1, x_2, x_3, \dots, x_n$ , is a rule that assigns a unique real number.

$w = f(x_1, x_2, x_3, \dots, x_n)$  in some set  $D$  of  $n$  dimensional space.

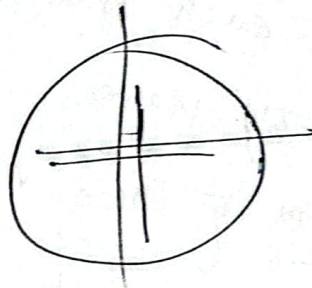
## Analytical Geometry



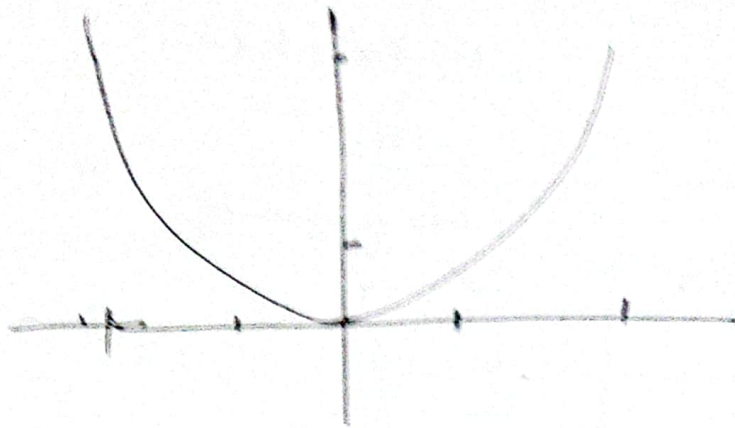
$$(x-a)^2 + (y-b)^2 = r^2$$



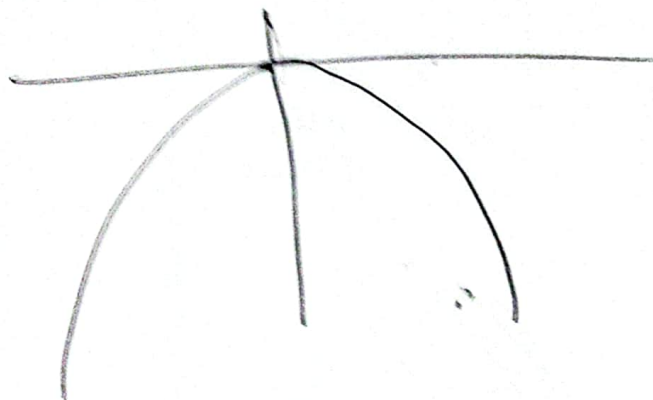
$$x^2 + y^2 = a^2$$



$$x^2 + y^2 \leq a^2$$



⑦



$$y = f(x) = ax^2 + bx + c$$

opening upward if  $a > 0$

opening downward if  $a < 0$

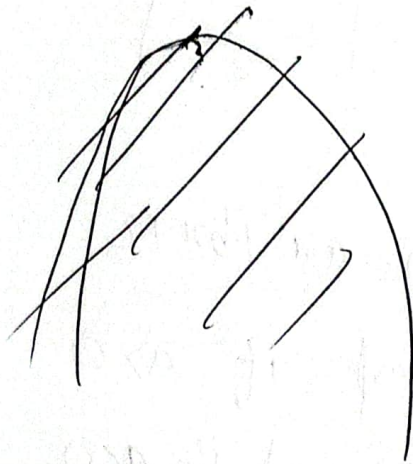
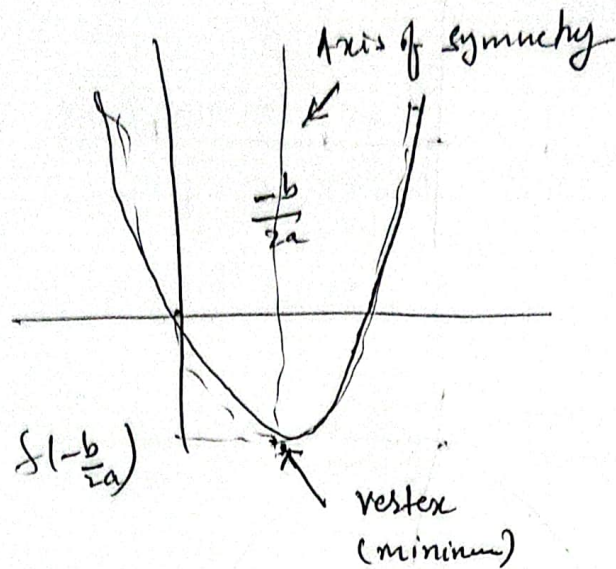
x co-ordinate of the vertex is given by  
 $x_0 = -b/2a$

so the y coordinate of the vertex is  
 $y_0 = f(x_0)$

axis of symmetry is  $x = x_0$

$$y = ax^2 + bx + c$$

$$(a > 0)$$



Ex:  $y = -x^2 + 4x$

since  $a = -1 < 0$ , parabola is opening downward.  
 vertex occurs at  $x = -b/2a = (-4)/2(-1) = 2$

Axis of symmetry is the vertical line  $x = 2$ .  
 the y coordinate of the vertex is

$$y = -(2)^2 + 4(2) = 4$$

Hence vertex is  $V(2, 4)$

$$y = -x^2 + 4x$$

Q

The zeros of the parabola (i.e. the point where the parabola meets  $x$ -axis) are the solutions to

$$-x^2 + 4x = 0 \quad \text{so } x = 0 \text{ and } x = 4$$

therefore  $(0,0)$  and  $(4,0)$  lie on the parabola

Also  $(1,3)$  and  $(3,3)$  lie on the parabola.

